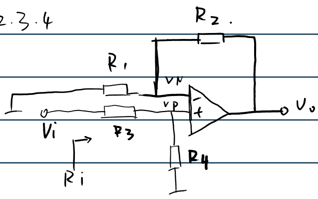
模电作业.2

2025年9月5日　22:53　　　　自动化2406 迟泰炎U202414215

2.3.4



(1) $\begin{cases} \dfrac{V_i - V_P}{R_3} = \dfrac{V_P}{R_4} \\ \dfrac{V_o - V_P}{R_2} = \dfrac{V_P}{R_1} \end{cases}$

$\therefore A_v = \dfrac{V_o}{V_i} = \dfrac{\frac{R_1+R_2}{R_1}}{\frac{R_3+R_4}{R_4}} = \dfrac{R_4(R_1+R_2)}{R_1(R_3+R_4)}$

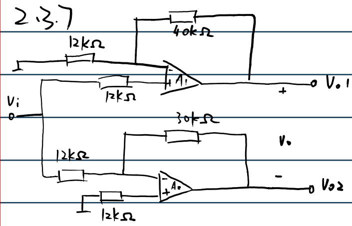
$R_i = \dfrac{V_i}{I_i} = \dfrac{\frac{R_3+R_4}{R_4}V_P}{\frac{V_P}{R_4}} = R_3+R_4$

(2). $\pm10\pm2$

$\therefore$ $\pm12V$以上

(3). $R = \dfrac{U}{I} = \dfrac{10}{(20-0.5)\times10^{-3}} \approx 513\Omega.$

2.3.7



(1). $V_i = 0.5\sin\omega t$

$V_{o1} = \left(1+\dfrac{40k}{12k}\right)V_i \approx 2.17\sin\omega t$

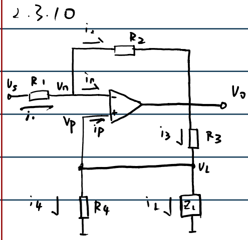
$V_{o2} = \left(-\dfrac{30k}{12k}\right)V_i = -1.25\sin\omega t$

$V_o = V_{o1} - V_{o2} = 3.417\sin\omega t.$

(2). $A_v = \dfrac{V_o}{V_i} = \dfrac{3.417\sin\omega t}{0.5\sin\omega t} = 6.834.$

$R_i = 12\Omega$

2.3.10

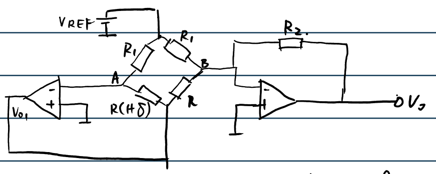


$\dfrac{R_2}{R_1R_3} = \dfrac{1}{R_4}$

$\begin{cases} V_n = V_p = V_L = i_L \cdot Z_L. \\ \dfrac{V_s - V_n}{R_1} = \dfrac{V_n - V_o}{R_2} \\ \dfrac{V_o - V_L}{R_3} = i_L + \dfrac{V_L}{R_4} \end{cases}$

$\therefore \dfrac{i_L}{V_s} = -\dfrac{1}{R_4}$

2.3.11



$V_o \sim \delta$

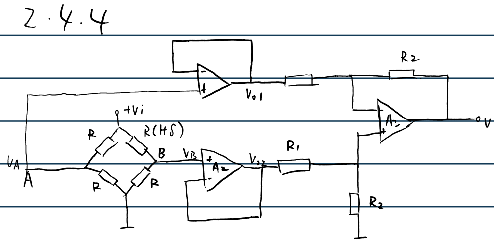
$U_A = U_B = 0$

对A列KCL. $\dfrac{V_{o1}-0}{R(1+\delta)} = \dfrac{0-V_{REF}}{R_1}$

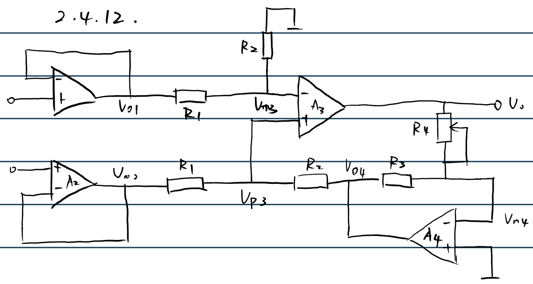
对B列KCL $\dfrac{V_o-0}{R_2} + \dfrac{V_{REF}-0}{R_1} = \dfrac{0-V_{o1}}{R}.$

联立得: $V_o = \dfrac{R_2}{R_1}\cdot V_{REF}\cdot\delta$

2.4.4



2.4.12.



推导 $A_v = \dfrac{V_o}{V_{i1}-V_{i2}}$

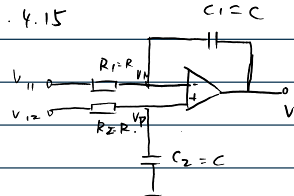
$V_{o1} = V_{i1}\quad V_{o2} = V_{i2}\quad V_{n3} = V_{p3}$

$\begin{cases} V_{o4} = -\dfrac{R_3}{R_4}V_o \\ \dfrac{V_{o1}-V_{n3}}{R_1} = \dfrac{V_{n3}}{R_2} \\ \dfrac{V_{o2}-V_{p3}}{R_1} = \dfrac{V_{p3}-V_{o4}}{R_2}. \end{cases}$

$R_2V_{o1} - R_2V_{o2} = V_o\dfrac{-R_1R_3}{R_4}.$

$\therefore A_v = \dfrac{V_o}{V_{i1}-V_{i2}} = -\dfrac{R_2R_4}{R_1R_3}.$

2.4.15



(1). $V_p(s) = \dfrac{\frac{1}{sC}}{R+\frac{1}{sC}}V_{i2}(s).$

$V_o'(s) = \left[1+\dfrac{\frac{1}{sC}}{R}\right]V_p(s)$

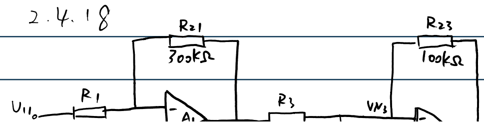
$= \dfrac{1}{sCR}V_{i2}(s) = \dfrac{1}{RC}\int_0^t v_{i2}dt$

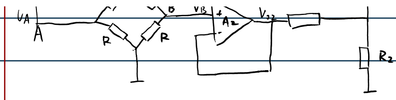
(2). $V_o''(s) = -\dfrac{1}{sRC}V_{i1}(s)$

$V_o'' = -\dfrac{1}{RC}\int_0^t v_{i1}dt$

(3). $V_o = V_o' + V_o'' = \dfrac{1}{RC}\int_0^t (V_{i2}-V_{i1})dt$

2.4.18





(1). $V_{o1} = V_A = \frac{V_i}{2}$ $\quad V_{o2} = \frac{R}{2R+\delta R}\cdot V_i = \frac{1}{2+\delta}V_i$

$V_o = \frac{R_2}{R_1}(V_{o2}-V_{o1}) = \frac{R_2}{R_1}\cdot\left(\frac{1}{2+\delta}-\frac{1}{2}\right)V_i = \frac{R_2 V_i}{R_1}\cdot\frac{-\delta}{4+2\delta}$

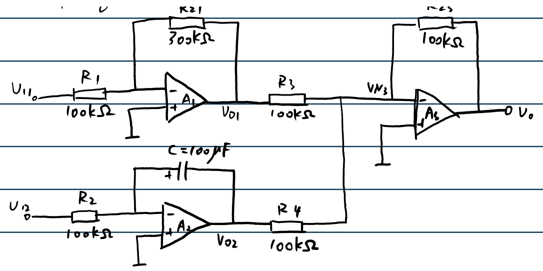
(2) $V_A = \frac{1}{2}V_i = 3.75V$

$V_B = \frac{1}{2+\delta}V_i = 3.73V$

$V_{AB} = 0.02V$.

$V_i = \frac{R_2}{R_1}V_i\frac{-\delta}{4+2\delta} = -0.02\frac{R_2}{R_1}$

(3). 降压.



(1). $\frac{V_{I1}-0}{100k} = \frac{0-V_{o1}}{300k}$ $\quad \frac{V_{o1}-V_{N3}}{R_3}+\frac{V_{o2}-V_{N3}}{R_4} = \frac{V_{N3}-V_o}{R_{23}}$

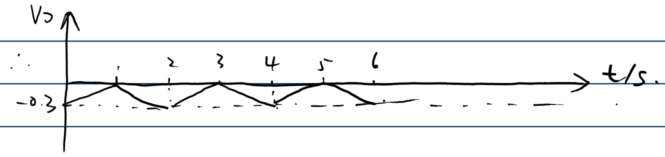
$V_{o1} = -3V_{I1}$. $\quad V_o = 3V_{I1}+\frac{1}{RC}\cdot\int V_{I2}dt$.

$0-V_N = \frac{1}{C}\int i\,dt$. $\quad = 3V_{I1}+0.1\int V_{I2}dt$.

$V_{o2} = -\frac{1}{RC}\cdot\int V_{I2}dt$.

(2). $V_{I1} = -0.1V$. $\quad 3V_{I1} = -0.3V$.

∴

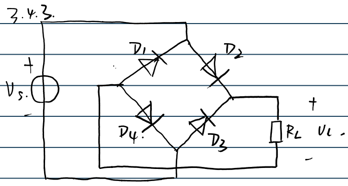


模电作业.3

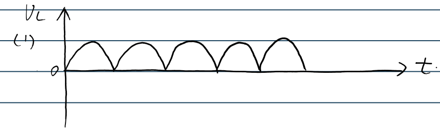
2025年9月20日 16:30

U202414215 迟泰炎 自动化2406.

3.4.3.

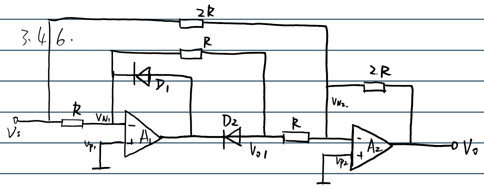


(1)



(2) $V_m = 220\sqrt{2}$ V.

3.4.6.



(1). 当$V_S > 0$时. D_1截止 D_2导通.

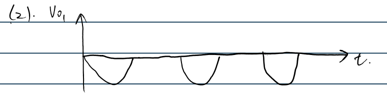
$V_{O1} = -V_S$.

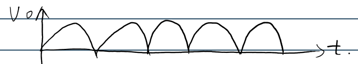
$V_o = -\frac{2R}{2R}V_S - \frac{2R}{R}V_{O1} = -V_S - 2(-V_S) = V_S \quad (V_S > 0)$.

$V_S \le 0$时. $V_{O1} = 0$.

$\therefore V_o = -\frac{2R}{2R}V_S = -V_S$

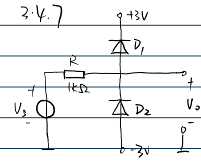
(2).





(3) 3.4.3 V_L始终为0

3.4.7

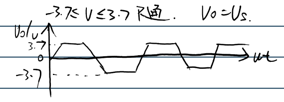


$V_S = 6\sin\omega t$.

$V > 3.7$ D_1通.

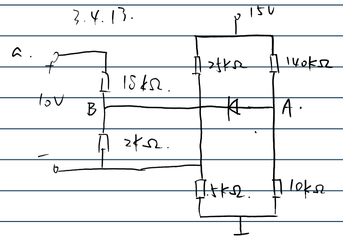
$V < -3.7$ D_2通.

$-3.7 \le V \le 3.7$ R通. $V_O = V_S$.



3.4.13.

a.

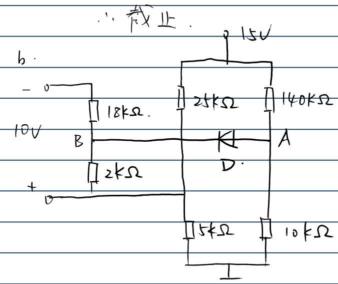


$V_A = 1V$.

$V_B = \frac{2}{18+2} \times 10 + \frac{5}{25+5} \times 15 = 3.5V$

$\therefore$ 截止.

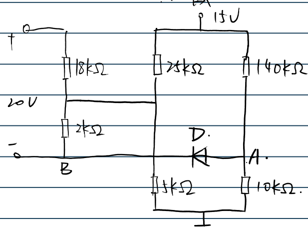
b.



$VA = \frac{10k\Omega}{150k\Omega} \times 15 = 1V$.

$V_B = -\frac{2}{18+2} \times 10 + \frac{5}{25+5} \times 15 = 1.5V$

$\therefore$ 截止

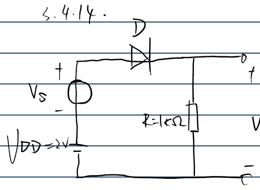


$V_A = \frac{10}{10+140} \times 15 = 1V$.

$V_B = \frac{5}{25+5} \times 15 - \frac{2}{18+2} \times 20 = 0.5V$ ✓

$\therefore$ 导通

3.4.14.



$V_S = 50\sin(2\pi \times 50t)$ mV.

(1) $\because V_D = 0.7V$.

$\therefore I = \frac{2-0.7}{1k} = 1.3 \times 10^{-3}$ A.

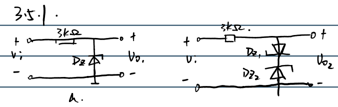
$V_O = I \cdot R = 1.3V$.

(2). $r_d = \frac{V_T}{I_D} = \frac{26 \times 10^{-3}}{1.3 \times 10^{-3}} = 20\Omega$.

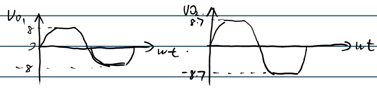
$\therefore i_d = \frac{V_S}{R+r} = \frac{50}{1020} \times 10^{-3} \approx 0.049V$

(3). $v = V_1 + V_2 = 1.3 \times 10^{-3} + 0.049\sin(2\pi \times 50t)$ V.

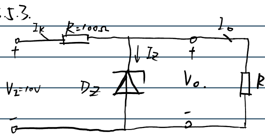
3.5.1.



$V_i = 15\sin\omega t$ V.



3.5.3.



$V_Z = 5V$

$I_Z \in [5, 50]$ mA.

(1) $I_R = \frac{V_I - V_Z}{R} = \frac{10-5}{100} = 0.05A$.

$I_R - I_{OM} > I_{Zmin}$.

$0.05 - I_{OM} > 5 \times 10^{-3}$

$\therefore I_{OM} < 0.045A$.

(2) $P_{OM} = I_{OM} \times V_Z = 0.225W$

(3). $P_{ZM} = V_Z \cdot I_{Rmax} = 5 \times 5 \times 10^{-2} = 0.25W$

$P_{RM} = (V_I - V_Z) I_{Rmax} = (10-5) \times 50 \times 10^{-3} = 0.25W$

模电作业.4

U202414215 迟泰炎 自动化2406

2025年10月8日 16:48

4.1.1 4.2.1 4.4.3 4.4.4 4.5.2 4.5.5 (4.7.1)

4.1.1.

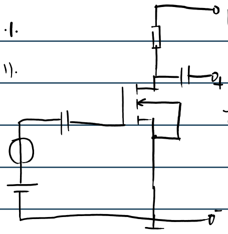
(1) N耗尽型. $V_{TN}=-3V$.

(2). P耗尽型. $V_{TP}=2V$.

(3). P增强型. $V_{TP}=-4V$.

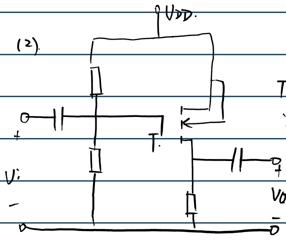
4.2.1.

(1).



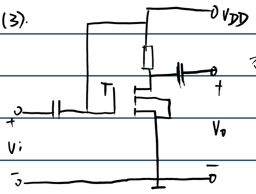
T为增强型NMOS.
C_{b1} 使得 V_{GG} 无法到栅极.
无交流放大.

(2).



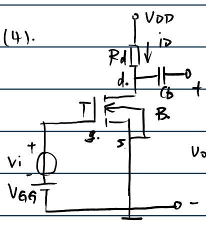
T为增强型PMOS.
源极电位 有交流放大作用.

(3).



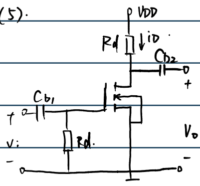
交流通路V_{DD}接地
无放大作用.

(4).



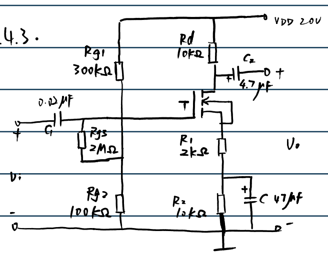
增强型NMOS.
V_{GG}. V_{DD} 可提供足够偏置
有交流放大作用.

(5).

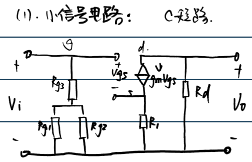


增强型NMOS
栅极绝缘.
无放大作用.

4.4.3.



(1). 小信号电路: C短路.

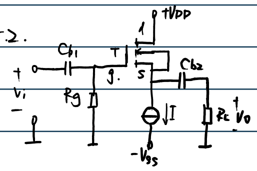


(2). $V_o=-g_mV_{gs}\cdot R_d$.

$V_i=V_{gs}+g_mV_{gs}\cdot R_1$.

$\therefore A_v=\frac{V_o}{V_i}=\frac{-g_mV_{gs}R_d}{\dots}=\frac{-g_mR_d}{1+g_mR_1}=\frac{-10}{1+2}=-3.3$

4.5.2.



$K_n=1mA\cdot V^{-2}$. $R_g=5\times10^5\Omega$
$V_{TN}=1.2V$. $R_L=4\times10^3\Omega$
$\lambda=0$. $I=1\times10^{-3}A$.
$V_{DD}=V_{SS}=5V$.

$I=I_{DQ}=K_n(V_{GSQ}-V_{TN})^2=(V_{GSQ}-1.2)^2\times10^{-3}$.

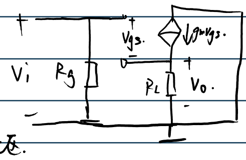
$(V_{GSQ}-1.2)^2=1$.

$V^2-2.4V+1.44=1$

$V^2-2.4V+0.44=0$.

$\therefore V=\frac{2.4\pm2}{2}$

$V_1=2.2V$ $V_2=0.2V$



$\therefore V_{GSQ}=2.2V$.

$V_{DSQ}=V_D-V_S=7.2V$.

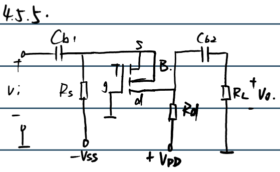
$V_{DSQ}>V_{GSQ}-V_{TN}>0$. ∴恒流区.

$\therefore g_m=2\sqrt{K_n\cdot I_{DQ}}=2\times10^{-3}S$.

$A=\frac{V_o}{V_i}=\frac{g_mV_{gs}R_L}{g_mV_{gs}R_L+V_{gs}}=\frac{g_mR_L}{g_mR_L+1}=\frac{8}{9}\approx0.89$.

$R_o=\frac{1}{g_m}=0.5k\Omega$.

4.5.5.



$V_{DD}=V_{SS}=5V$. $V_{TN}=1V$.
$R_s=10^4\Omega$. $\lambda=0$.
$R_d=5\times10^3\Omega$.
$R_L=5\times10^3\Omega$
$K_n=3\times10^{-3}A\cdot V^{-2}$

(1). $\begin{cases}I_{DQ}=K_n(V_{GSQ}-V_{TN})^2 & \cdots ① \\ -V_{SS}+I_{DQ}R_s+V_{GSQ}=0. & \cdots ②\end{cases}$

由②. $V_{GSQ}=5-10I_{DQ}$

$\therefore I_{DQ}=3\times(5-10I_{DQ}-1)^2$.

$3\times(16+100I_{DQ}^2-80I_{DQ})$.

$300I_{DQ}^2-241I_{DQ}+48=0$.

$I_{DQ}=\frac{241\pm\sqrt{481}}{600}\approx\frac{241\pm21.932}{600}$

$I_{DQ1}\approx0.438$ $I_{DQ2}\approx0.367$.

$\therefore V_{GSQ}=5-10I_{DQ}\approx1.33V$.

$\therefore V_S=0-U_{GSQ}=-1.33V$.

$\therefore V_{DSQ}=V_D-V_S=V_{DD}-I_{DQ}R_d-V_S$

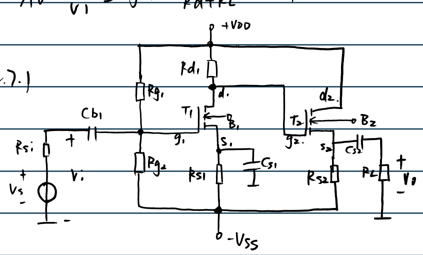
$\approx4.495V$.

$V_{DSQ}>V_{GSQ}-V_{TN}>0$.

(2). $g_m=2K_n\cdot(V_{GSQ}-U_{TN})\approx1.98\times10^{-3}S$.

(3). $A_v=\frac{V_o}{V_i}=g_m\cdot\frac{R_dR_L}{R_d+R_L}=1.98\times10^{-3}\times2.5\times10^3=4.95$.

4.7.1



$R_L=5k\Omega$. $K_{n1}=K_{n2}=1mA\cdot V^{-2}$
$R_{d1}=3.3k\Omega$. $V_{TN1}=V_{TN2}=1.3V$
$R_{s1}=20k\Omega$. $\lambda_1=\lambda_2=0$.
$R_{s2}=5k\Omega$.
$R_{g1}=500k\Omega$.
$R_{g2}=300k\Omega$
$V_{GS1Q}=V_{GS2Q}=3V$

(1)

$V_i = V_{gs} + g_m V_{gs} \cdot R_1$

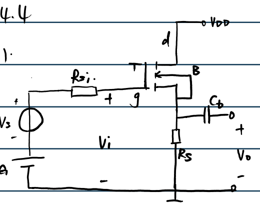
$\therefore A_v = \frac{V_o}{V_i} = \frac{-g_m V_{gs} R_d}{V_{gs} + g_m V_{gs} R_1} = \frac{-g_m R_d}{1+g_m R_1} = \frac{-10}{1+2} = -3.3$

(3). $R_i = R_{g3} + \frac{R_{g1} R_{g2}}{R_{g1}+R_{g2}} = 10^6 + \frac{3\times10^3\times10^3}{4\times10^3} = 2075\,k\Omega$.

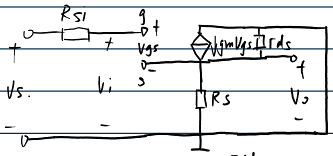
(4). ?

4.4.4

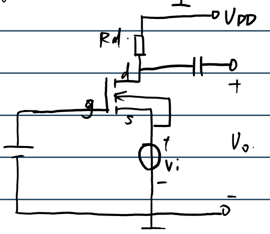
(1).



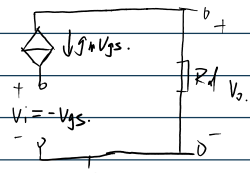
⇓



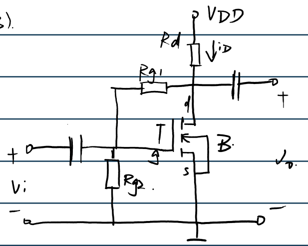
(2).



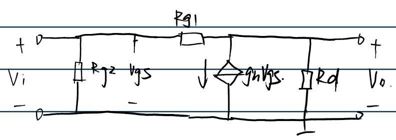
⇓

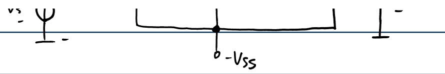


(3).



⇓





$V_{GS1Q} = V_{GS2Q} = 3V$

(1).

$g_{m1} = g_{m2} = 2K_n(V_{GS1Q} - V_{TN1}) = 3\times10^{-3}$ S.

$R_i = \frac{5\times3\times10^{3}}{5+3} = 187.5\times10^3\,\Omega$.

$R_o = R_{s2} // \frac{1}{g_m} // r_{ds2} = R_{s2} // \frac{1}{g_m}$

$= \frac{1}{\frac{1}{5\times10^3} + 3\times10^{-3}} = 312.5\,\Omega$.

(2). $A_v = \frac{V_o}{V_i} = -\frac{g_{m1} g_{m2} R_{d1} \frac{R_L R_{s2}}{R_L + R_{s2}}}{1 + g_{m2} \frac{R_{s2} R_L}{R_{s2} + R_L}} \approx -8.735$.

$\therefore A_{vs} = A_v \cdot \frac{R_i}{R_i + R_{si}} = -8.735 \times 0.9056 \approx -7.893$.

模电作业.5

5.1.1 5.2.1 5.2.7 5.2.9 5.2.12 5.4.4

2025年10月29日 23:47

5.1.1

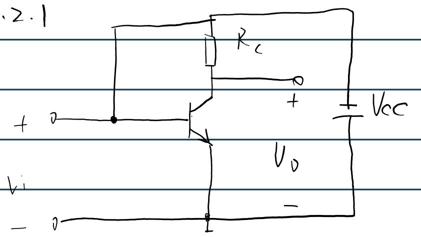
易知 C为基极；

$V_C - V_B = -0.2 < 0$.

∴ B为射极

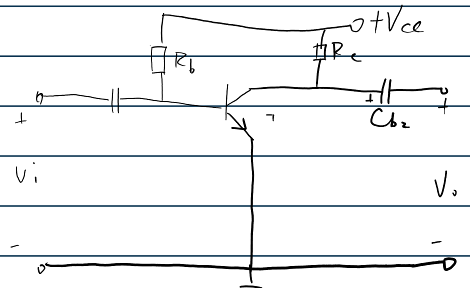
∴ A为集电极；PNP

5.2.1

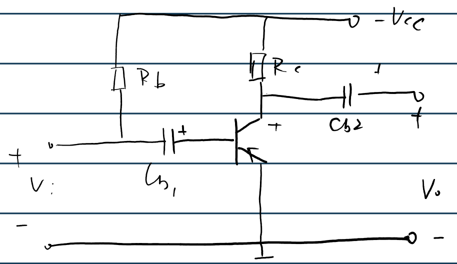


直接交流时 V_i 被短路

∴无法放大.

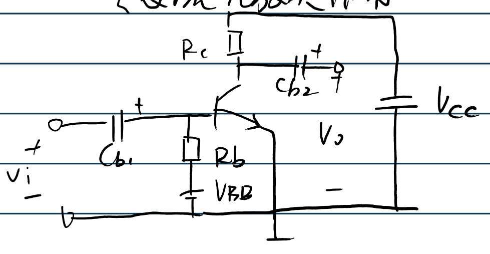


有电流放大



C_{b1} 阻断直流通路

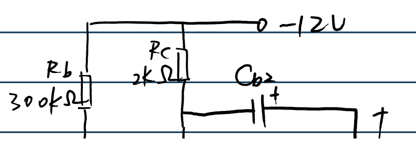
∴交流无放大作用



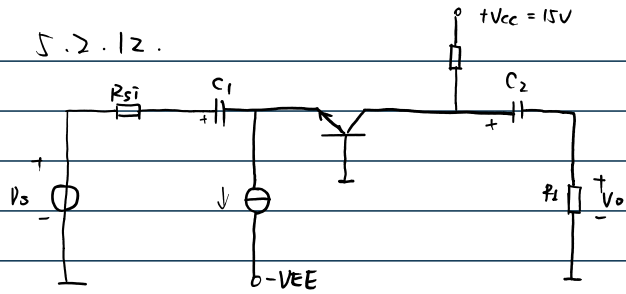
∴ C_{b1} 阻断基极

∴无放大

5.2.7



5.2.12.



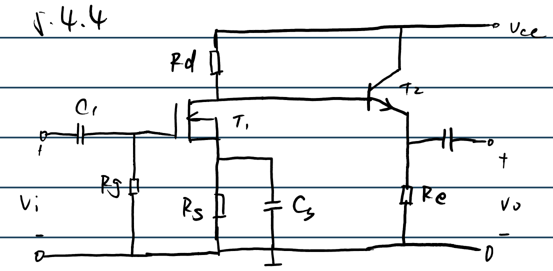
$r_{be} = 200 + (1+\beta)\frac{26}{I_{EQ}} \approx 2.8k\Omega$.

$A_v = \frac{\beta \cdot R_L}{r_{be}} \approx 268$

∴ $R_i = \frac{r_{be}}{1+\beta} \approx 28\Omega$

$R_o \approx R_c = 7.5k\Omega$

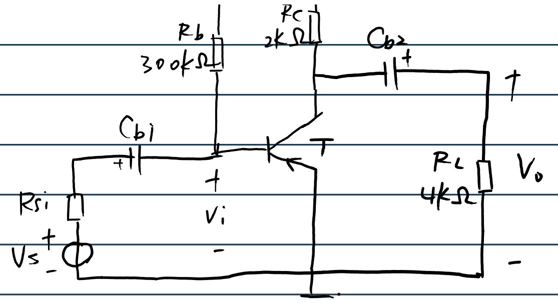
5.4.4



$A_{v1} = -g_{m1} \cdot R_{L1}$ $A_{v2} = \frac{(1+\beta)R_e}{(1+\beta)R_e + r_{be}}$

$A_v = A_{v1}A_{v2} = -\frac{g_{m1}(1+\beta)R_eR_d}{r_{be}+R_d+(1+\beta)R_e}$

$R_i = R_g$ $R_o = \frac{(R_d + r_{be})R_e}{R_e(1+\beta)+R_d+r_{be}}$

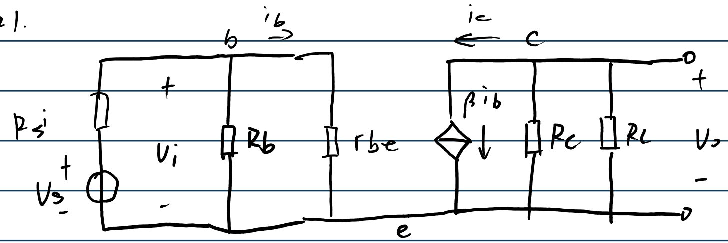


(1) $I_{BQ} = \frac{-V_{BEQ}+V_{CC}}{R_b} \approx 40\mu A$.

$\therefore I_{CQ} = \beta I_{BQ} \approx 4mA$

$V_{CEQ} \approx -12 + R_c \cdot I_{CQ} \approx -4V$.

(2).



(3). $r_{be} = 200 + (\beta+1)\frac{V_T}{I_{CQ}} \approx 857\Omega$.

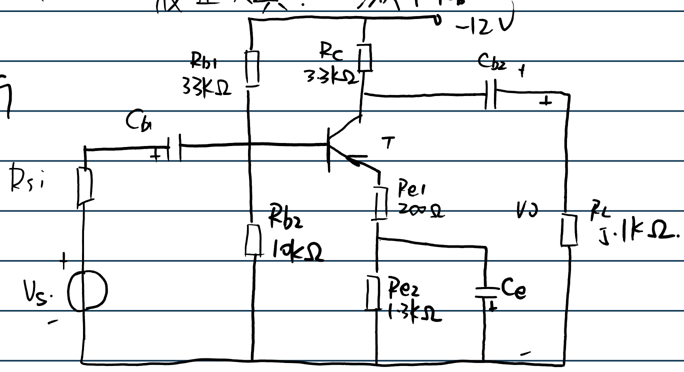
$A_v = \frac{V_o}{V_i} \approx -155.7$

$R_i = \frac{R_b r_{be}}{R_b + r_{be}} \approx 855\Omega$.

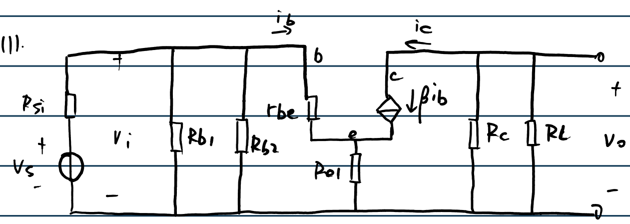
$R_o \approx R_c = 2k\Omega$.

(4). 截止失真. 减小 R_b.

5.2.9



(1).



(2). $V_{BQ} = \frac{R_{b2}}{R_{b1}+R_{b2}} \cdot (-12) \approx -2.8V$.

$I_{EQ} = \frac{-V_{EQ}}{R_{e1}+R_{e2}} = 1.4 \times 10^{-3} A$.

$\because r_{be} = 1147\ \Omega$

$R_i = 4.6k\Omega$. $R_o \approx R_C = 3.3k\Omega$

(3). $A_{vs} = \frac{V_o}{V_s} \approx -7.8$.

$V_o = A_{vs} \cdot V_s \approx -117 \sin\omega t \times 10^{-3} V$.

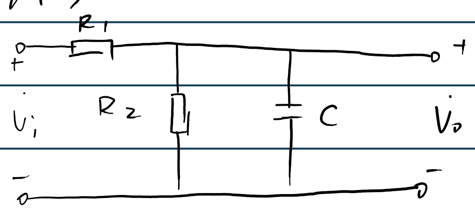
(4).

自动化2406 迟泰炎 U202414215

2025年11月8日 20:03

6.1.3. 6.3.1. 6.3.2.

6.1.3.



(1). 低通路。

(2). $\dot A_v = \frac{\dot V_o}{\dot V_i} = \frac{R_2}{R_1+R_2}\cdot\frac{1}{1+j\omega C\frac{R_1R_2}{R_1+R_2}} = \frac{R_2}{R_1+R_2}\cdot\frac{1}{1+j\frac{f}{f_H}}$

当 $f=0$ 取得最大值 $\frac{10}{11}$

(3). $f_H = \frac{1}{2\pi C\frac{R_1R_2}{R_1+R_2}} = 58.4\,MHz$.

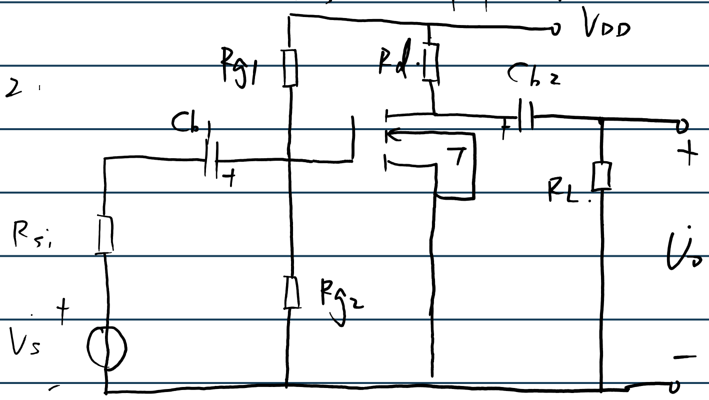
6.3.1

(1). $C = C_{gs} + C_{gd}(1+g_mR_L) = 3.1\times10^{-9}F$.

(2). $f_H = \frac{1}{2\pi C R_{si}} = \frac{1}{2\times3.14\times3.1\times10^{-9}\times R_{si}} > 10^7 Hz$

$\therefore R_{si} < 5.1\,k\Omega$.

6.3.2.



$V_{GSQ} = \frac{R_{g2}}{R_{g1}+R_{g2}}\cdot V_{DD} \approx 1.7V$.

$\therefore V_{DSQ} \approx 3V.\ I_{DQ} \approx 0.39\,mA$ 在恒流区.

$g_m = 2K_n(V_{GSQ}-V_{TN})$.

$= 1.12\,mS$.

$R_L' = 7.57k\Omega$.

$\therefore \dot{A}_{VSM} = -g_m R_L' \cdot \frac{R_{g1} // R_{g2}}{R_{si} + R_{g1} // R_{g2}} \approx -2.65$.

$\therefore C = C_{gs} + C_{gd}(1 + g_m R_L') \approx 2.54 \times 10^{-9} F$.

$R_{si}' \approx 1.86k\Omega$.

$f_H = \frac{1}{2\pi C R_{si}'} = 33.7 MHz$

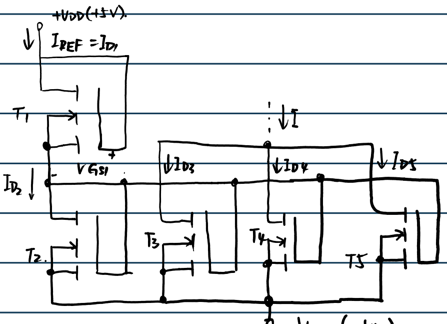
模电作业.7

迟泰炎 U202414215 自动化2406

2025年11月13日 15:09

7.1.4　7.2.1　7.5.1　7.5.3　7.5.4　7.5.6.

7.1.4



$I_{REF} = I_{D1} = I_{D1}$.

$\therefore \begin{cases} K_{n1}(V_{GS1}-V_{TN})^2 = K_{n2}(V_{GS2}-V_{TN})^2 \\ V_{DD}+V_{SS} = V_{GS1}+V_{GS2} \end{cases}$

$\therefore \begin{cases} 10 = V_{GS1}+V_{GS2} \\ 0.1(V_{GS1}-2)^2 = 0.25(V_{GS2}-2)^2 \end{cases}$

$\therefore 0.15V_{GS2}^2 + 0.6V_{GS2} - 5.4 = 0$.

$V_{GS2} = \dfrac{-0.6+\sqrt{0.36+0.6\times5.4}}{0.3}$.

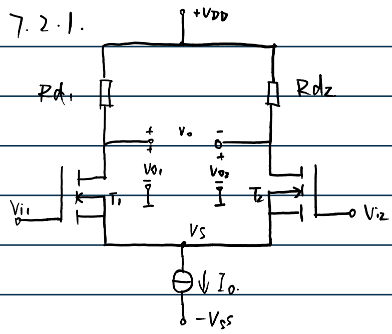
$\therefore V_{GS2} \approx 4.3V$.

$\therefore V_{GS1} \approx 5.7V$

$\therefore I_{REF} = 0.1(V_{GS1}-V_{TN})^2 \approx 1.37mA$.

$I = 3I_{REF} \approx 4.11mA$.

7.2.1.



$V_{GS1}-V_{TN} = \sqrt{\dfrac{I_{D1}}{K_n}} \approx 0.26V$

$V_{D1} = V_{DD} - I_{D1}R_{d1} = 4V$.

$V_S = -V_{GS1}$.

$\therefore V_{DS1} = V_{D1}-V_S = 4+V_{AS1} = 4+0.26+V_{TN} > 0$ 恒成立.

$\therefore$ 位于恒流区.

$\therefore g_m = 2\sqrt{K_n I_{D1}} = 0.77ms$.

$\therefore A_{vd2} = \dfrac{V_{o2}}{V_{id}} = \dfrac{1}{2}g_m R_d \approx 3.9$.

$A_{vc2} = \dfrac{V_{o2}}{V_{ic}} \approx -\dfrac{R_d}{2r_o} = -0.05$.

$K_{CMR} = \left|\dfrac{A_{vd2}}{A_{vc2}}\right| \approx 78$.

7.5.1

$V_{IO} = 1mV$.

$I_{IB} = 50nA$

7.5.3.

(1) $BW = \dfrac{f_T}{A_V} = \dfrac{5MHz}{10} = 500kHz$.

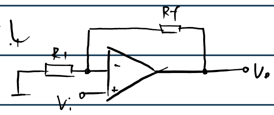
$BW_P = \dfrac{S_R}{2\pi V_{om}} \approx 1.91\times10^5 Hz$.

(2). $V_{om} = \dfrac{S_R}{5\times2\pi\times BW_P} \approx 2V$.

$A_V = \dfrac{f_T}{5\times BW_P} \approx 5.24$.

$\therefore V_{im} = \dfrac{V_{om}}{A_V} \approx 0.38V$.

7.5.4

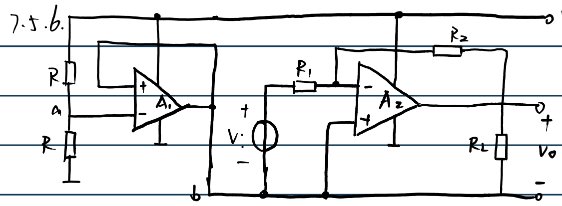


(1) $BW = \dfrac{f_T}{A_V} = 400MHz$.

$V_{om} = \dfrac{S_R}{2\pi\cdot BW} = 0.4V$.

(2) $BW_P = \dfrac{S_R}{2\pi\cdot V_{om}} \approx 159MHz$.

7.5.6.



(1) 对 A_2: $\dfrac{V_i}{R_1} = \dfrac{-V_o}{R_2}$

$A_V = -\dfrac{R_2}{R_1}$.

(2). $f_T \geq BW|A_V| = 2MHz$.

$\therefore S_R \geq 2\pi V_{om}\cdot f_{max} = 1.256V\cdot\mu s^{-1}$

(3). $i_{Rf} = \dfrac{V_{om}}{R_2} = 0.1mA$.

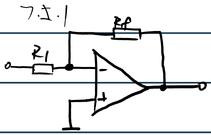
$\therefore i_{Lmax} = 10mA$.

$\therefore i_{omax} = i_{Lmax}+i_{Rf} = 10.1mA$.

(4). 不可.

连接可使得 A, 输出电阻逼近 0. 接近理想电压源.

7.5.1



$V_{IO}=1mV.$

$I_{IB}=80nA$

$I_{IO}=20nA$

(1). $V=\left(\frac{R_f}{R_1}+1\right)\left[V_{IO}+I_{IB}\cdot\frac{R_1R_f}{R_1+R_f}+\frac{1}{2}I_{IO}\frac{R_1R_f}{R_1+R_f}\right]\approx 11.9mV.$

(2). $V=\left(\frac{R_f}{R_1}+1\right)\left[V_{IO}+I_{IB}\frac{R_1R_f}{R_1+R_f}+\frac{1}{2}I_{IO}\frac{R_1R_f}{R_1+R_f}\right]\approx 100.98mV$

(3). $V_1=\left(\frac{R_f}{R_1}+1\right)V_{IO}+I_{IO}R_f=11.2mV.$

$V_2=\left(\frac{R_f}{R_1}+1\right)V_{IO}+I_{IO}R_f=31mV.$

∴小电阻时，平衡电阻对于减小误差作用更不明显.

模电作业.8

自动化2406 迟泰炎 U202414215

2025年11月22日 18:59

8.1.1. 8.1.2. 8.3.5 8.4.1 8.4.5 8.5.2.

8.1.1.

a. R_1 R_2 组成反馈网络. V_o 在R_1压降为反馈电压.
负反馈，直流反馈和交流反馈.

b. $A_2 \cdot R_3$ 组成反馈通路. A_2电压跟随器.
负反馈，直流反馈和交流反馈.

c. R_6 组成反馈网络 i_o在R_6压降为反馈电压
负反馈. 直流反馈和交流反馈.

d. R_1 R_2 组成反馈网络.
负反馈. 直流反馈和交流反馈.

e. R_f 组成反馈网络.
负反馈. 直流反馈和交流反馈.

f. R_{e1} 级间反馈网络.
负反馈. 直流反馈和交流反馈

8.1.2.

a. $R_1 R_2$ 串联负反馈

b. $A_2 R_3$ 并联负反馈

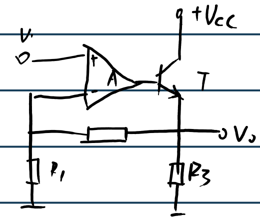
c. R_6 串联负反馈.

d. $R_1 R_2$ 并联负反馈.

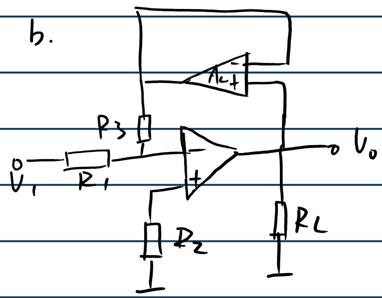
e. R_f. 并联负反馈

f. R_{e1} 串联负反馈.

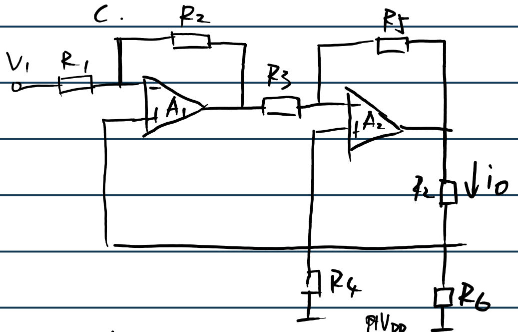
a.



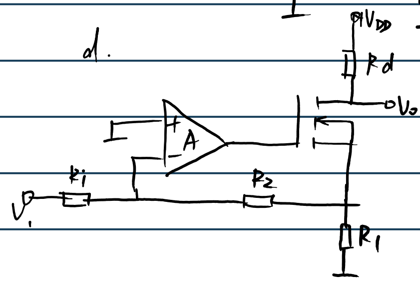
b.



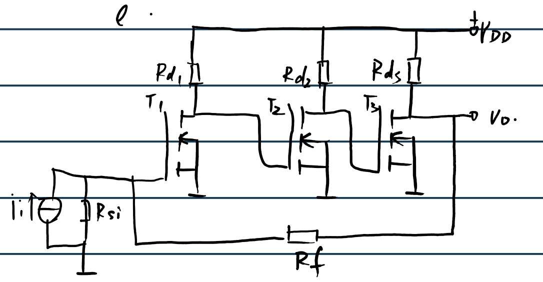
c.



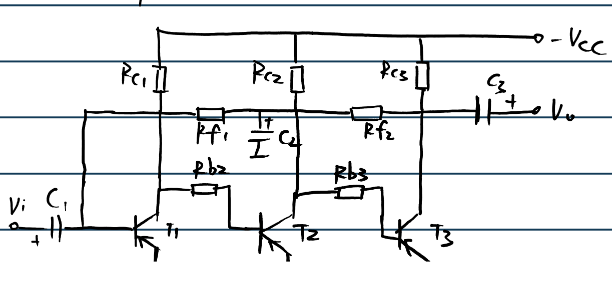
d.

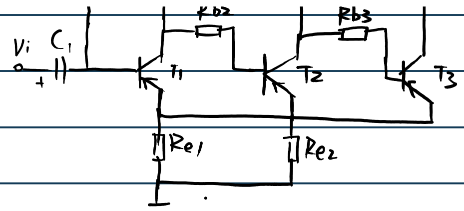


e.

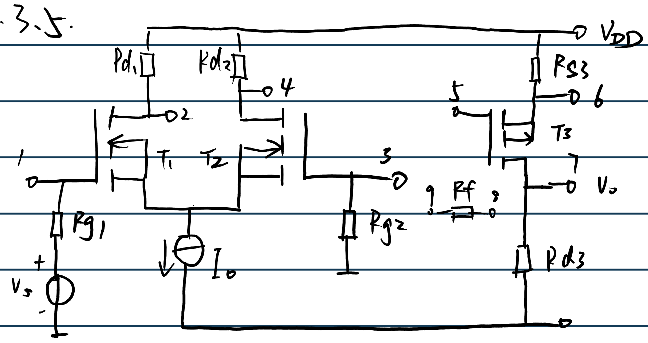


f.





8.3.5.



(1). 6−8 9−1 5−2.

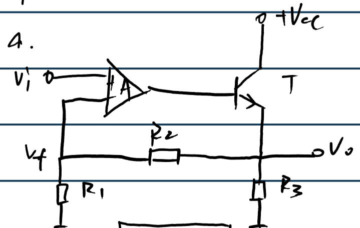
(2). 7−8 3−9. 2−5

(3). 7−8. 1−9. 4−5

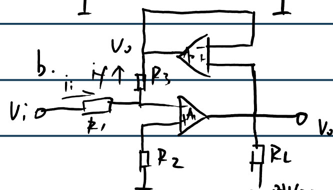
(4). 6−8 3−9. 4−5

8.4.1

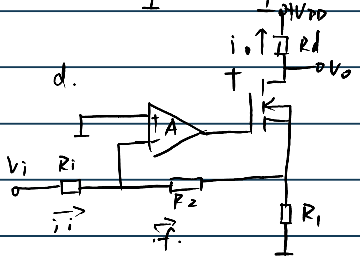
a.



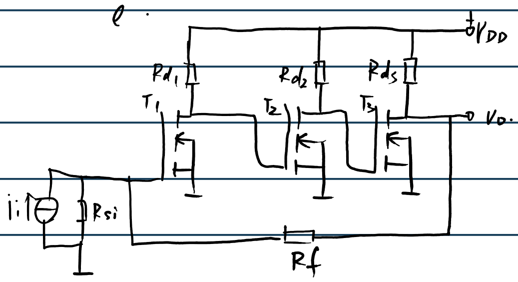
b.



d.



e.



f.

a. $V_i \approx V_f = \frac{R_1}{R_1+R_2} V_o$

$\therefore A_{vf} = \frac{V_o}{V_i} \approx 1 + \frac{R_2}{R_1}$

b. $i_i = i_f = -\frac{V_o}{R_3}$.

$A_{rf} = \frac{V_o}{i_i} \approx -R_3$

$A_{vf} = \frac{V_o}{V_i} = \frac{V_o}{i_i R_1} \approx -\frac{R_3}{R_1}$

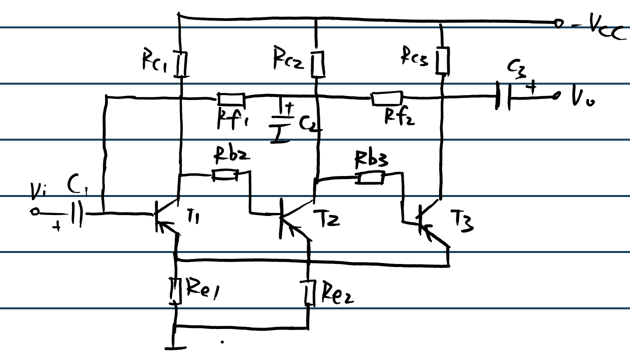
a. $F_i = \frac{i_f}{i_o} \approx \frac{R_1}{R_1+R_2}$

$A_{if} = \frac{i_o}{i_f} \approx 1 + \frac{R_2}{R_1}$.

e. $A_{rf} = \frac{V_o}{i_i} = -\frac{i_f R_f}{i_i} \approx -R_f$.

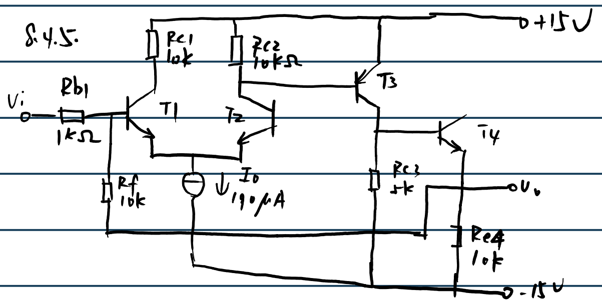
$A_{vf} = \frac{V_o}{V_s} \approx -\frac{R_f}{R_{si}}$

f. $A_{gf} = \frac{I_o}{V_i} = \frac{1}{R_{e1}}$.



$$A_{vf}=\frac{V_o}{V_i}\approx-\frac{R_{c3}R_{f2}}{R_{e1}(R_{c3}+R_{f2})}$$

8.4.5.



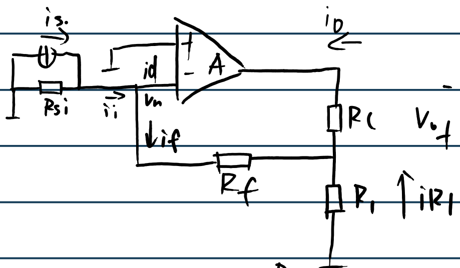
(1). $\frac{V_i}{R_1}\approx\frac{-V_o}{R_f}$　　$V_{b1}=V_{b2}=0$
$i_{b1}=0$

$\therefore A_{vf}=-10$.

(2). T_2基极用R_{b2}接到输入端.
T_1基极用 R_{b1} 接地.

(3). $A_{vf}\approx1+\frac{R_f}{R_{b1}}\approx11$.

8.5.2.



$i_i\approx i_f\approx\frac{R_1}{R_1+R_f}i_o$.

$\therefore A_{if}=10$.

$\therefore R_f=9R_1$.

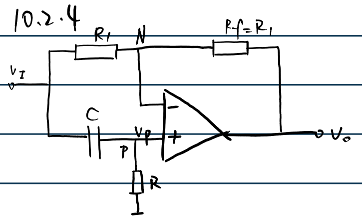
模电作业.9

自动化2406 迟泰炎 U202414215

2025年12月4日 19:29

10.8.1 10.8.6 10.8.8

10.2.4



(1) $\begin{cases} \dfrac{\dot V_i(j\omega)-\dot V_n(j\omega)}{R_1}=\dfrac{\dot V_n(j\omega)-\dot V_o(j\omega)}{R_1} \\ \dot V_n(j\omega)=\dot V_p(j\omega) \\ \dfrac{\dot V_i(j\omega)-\dot V_p(j\omega)}{\frac{1}{j\omega C}}=\dfrac{\dot V_p(j\omega)}{R} \end{cases}$

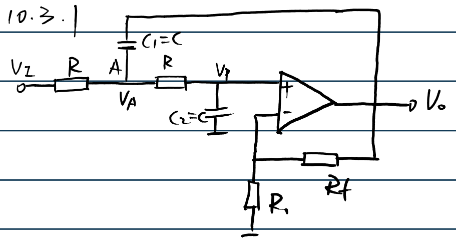
$\therefore \dfrac{\dot V_i(j\omega)}{1+j\omega RC}=\dfrac{\dot V_i(j\omega)+\dot V_o(j\omega)}{2j\omega RC}$

$\therefore \dot A(j\omega)=-\dfrac{1-j\omega RC}{1+j\omega RC}$

(2) $|\dot A_V(j\omega)|=\sqrt{\dfrac{1+(\omega CR)^2}{1+(\omega CR)^2}}=1$

$\varphi=-\pi-2\arctan(\omega RC)$

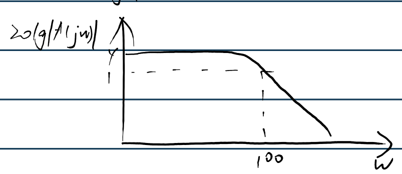
10.3.1



$A(s)=\dfrac{V_o(s)}{V_i(s)}=\dfrac{A_{vf}}{1+(sCR)^2+(3-A_{vf})sCR}=\dfrac{A_0}{1+(\frac{s}{\omega_c})^2+\frac{1}{Q}(\frac{s}{\omega_c})}$

$\therefore \omega_c=\dfrac{1}{RC}=100\,rad\,s^{-1}$ $\quad Q=\dfrac{1}{3-A_{vf}}=0.707$

$20\lg|A_0|=\cancel{20}\ 4dB$



10.6.1

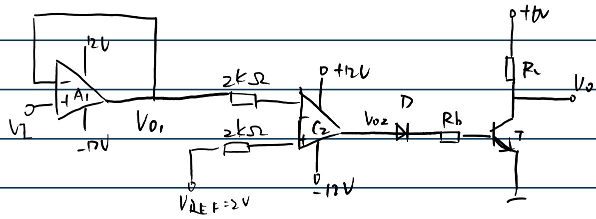
(1). 当T_1栅极为正，则T_2射极为负 $\varphi_a=180°$，反馈到T_1栅极为负

$\therefore \varphi_a+\varphi_f\neq 0$ 且$\neq 360°$，故不能振荡

(2) P为正，V_o为正，$\varphi_a=0$或$360°$. 在$\omega=\omega_0=\frac{1}{RC}$时 $\varphi_f=0°$.

$\therefore$经RC串并联网络到同相端信号也为正. $\varphi_a+\varphi_f=0$或$360°$. 可振荡

10.8.1



(1). $I_B=\dfrac{V_{o2}-V_{BE}}{R_b}\approx 222\mu A$

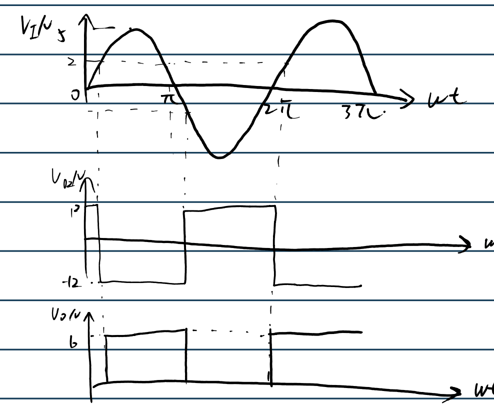
$I_{BS}=\dfrac{I_{CS}}{\beta}=\dfrac{V_{CC}-V_{CES}}{\beta\cdot R_c}=22.5\mu A$.

$\therefore I_B>>I_{BS}$. $\therefore V_{CES}=0$. $V_o=0$.

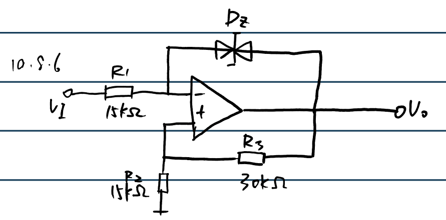
(2) $V_I=3$. $V_{o1}=3$

$\therefore V_{o1}>V_{REF}$. $V_{o2}\approx -12V$. $V_o=6V$.

(3). $V_I=5\sin\omega t$



10.8.6

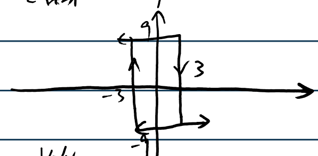


(1) $F=\dfrac{V_P}{V_o}=\dfrac{R_2}{R_2+R_3}=\dfrac{1}{3}$.

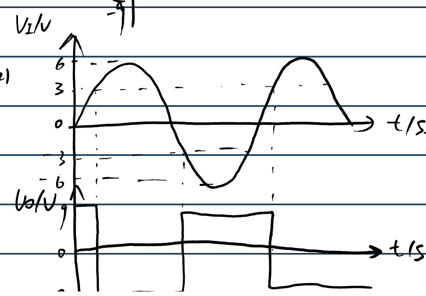
$V_N=V_P=\dfrac{V_o}{3}$

$\therefore V_o=\dfrac{V_o}{3}+V_Z=9V$.

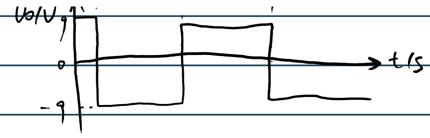
$\therefore D_Z$击穿. $V_o=-9V$.



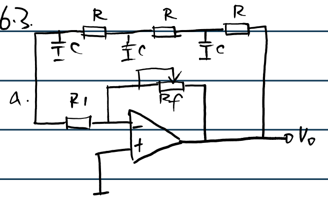
(2)

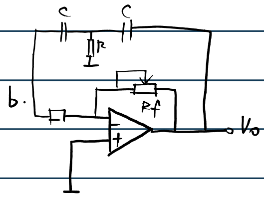


∴ 经RC串并联网络到同相端信号也为正. $\varphi_a+\varphi_f=0$ 或 $360°$. 可振荡



10.6.3.



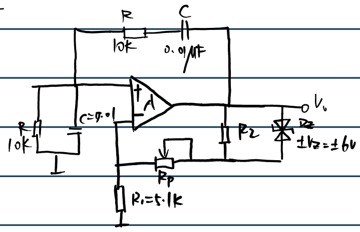


a. 运算放大器构成反相比例放大；输入输出相位差180°.

三阶RC可 270°. ∴存在 f_0 使得 $\varphi_a+\varphi_f=360°$ 可以振荡.

b. 二阶RC可 180°, $180°+180°=360°$ 因为二阶RC无法达到 180°. 所以不振荡

10.6.5



(1). 如图

(2). $|\dot{A}\dot{F}|>1$.

$\omega=\omega_0$, $|\dot{F}|=\frac{1}{3}$.

$\therefore\ 1+\frac{R_P+R_2}{R_1}>3$.

$\therefore R_P+R_2>10.2k\Omega$.

(3). $f=\frac{1}{2\pi RC}=1592Hz$.

(4). 稳定时有 $A_V=3$ $F_V=\frac{1}{3}$.

$\omega=\omega_0=\frac{1}{RC}$

$V_N=V_P=\frac{1}{3}V_{om}$.

$\therefore V_{RP}+V_Z=\frac{2}{3}V_{om}$.

$\therefore V_{RP}+V_Z=\frac{2}{3}V_{om}$

$\because V_{RP}=\frac{V_N}{R_1}R_P$.

$\therefore V_{RP}=\frac{V_{om}R_P}{3R_1}$.

$\frac{V_{om}R_P}{3R_1}+V_Z=\frac{2}{3}V_{om}$.

$V_{om}=\frac{3R_1}{2R_1-R_P}\cdot V_Z$.

10.8.8

(1) A_1: RC桥式正弦波振荡电路.

A_2: 过零比较器

A_3: 积分电路.